

# THE POST-DISASTER INFORMATION FOG

## D-FUSE

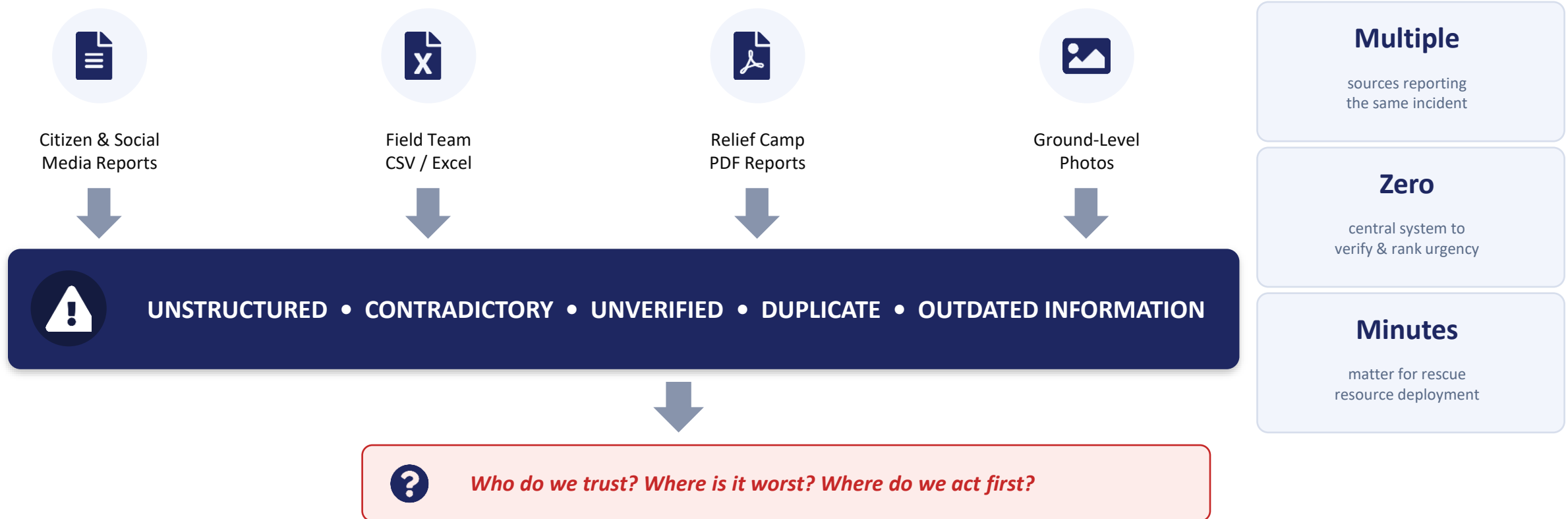
Dynamic Disaster Evidence Fusion & Uncertainty Engine

Team Name: TEAM CHANDRA

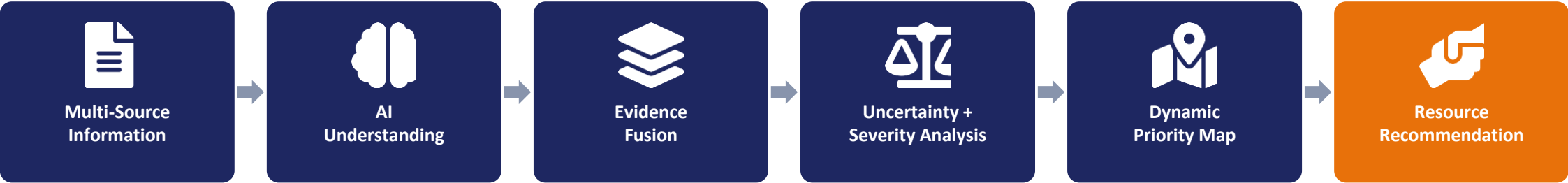
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# THE PROBLEM: POST-DISASTER INFORMATION FOG



# PROPOSED SOLUTION: THE D-FUSE PIPELINE



KEY OUTPUT

*“Where should responders act first?”*

## ACCEPTED INPUT TYPES



Text  
Reports



Excel /  
CSV



PDF  
Reports



Images



GPS /  
Location



Time-  
stamp



Source  
Info

# WHAT THE OUTPUT LOOKS LIKE



**VILLAGE A**  
Zone-E2 • Flood-affected

CRITICAL

|                     |                   |
|---------------------|-------------------|
| Severity Score      | 94 / 100          |
| Evidence Confidence | 91%               |
| People at Risk      | 350               |
| Road Access         | Blocked           |
| Trend (last 6 hrs)  | Rapidly Worsening |
| Independent Reports | 11 confirming     |

## PRIORITY LEGEND

|          |                    |
|----------|--------------------|
| CRITICAL | Act within minutes |
| HIGH     | Act within hours   |
| MEDIUM   | Monitor & plan     |
| LOW      | Routine follow-up  |

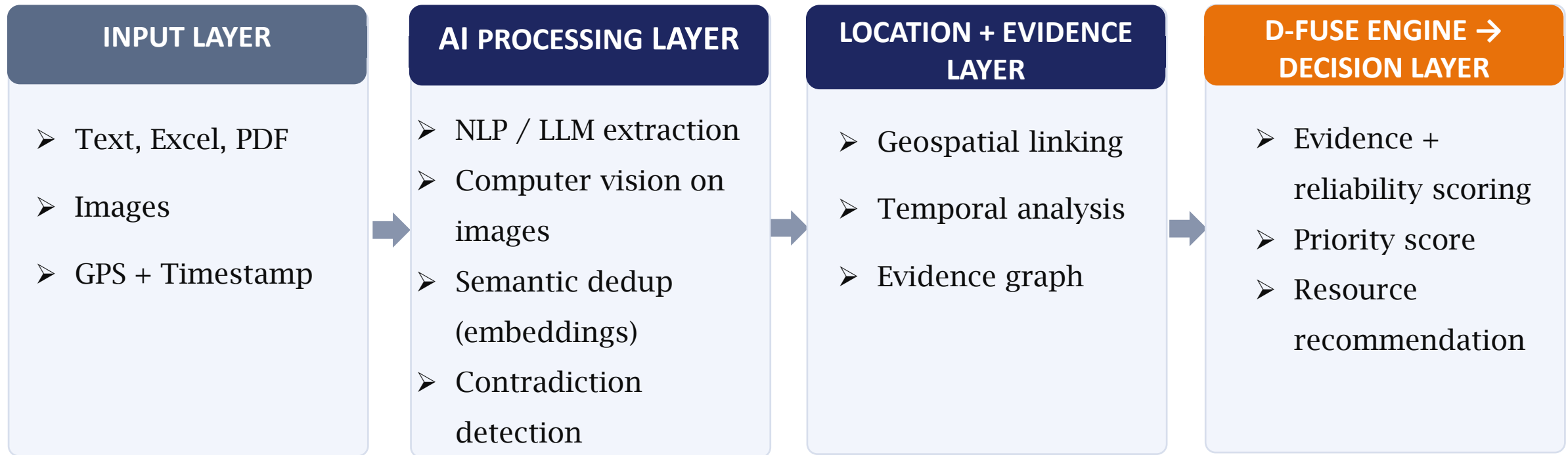
## RESOURCE-AWARE RECOMMENDATION

Deploy 2 rescue boats + 1 medical team to Village A now.

Village C (High) and Village D (Verify First) follow in the queue.

# TECHNICAL APPROACH: SYSTEM ARCHITECTURE

| THE POST-DISASTER INFORMATION FOG



## OUTPUT



D-FUSE

| THE POST-DISASTER INFORMATION FOG

# FEASIBILITY & VIABILITY

## TECHNOLOGY STACK

### Frontend

HTML/CSS/JS or React

### Backend

Python + Flask/FastAPI

### AI

LLM + Vision + Embeddings

### Database

PostgreSQL / Supabase

### Maps

Geospatial visualization

### Data

CSV/ Excel/ PDF/ Image



## DATA INGESTION

- ❖ Single report, bulk Excel/CSV, PDFs & images
- ❖ 1,000 reports → processed as one evidence pool

## LOCATION SOURCING

- ❖ GPS, map-pin selection, text extraction, or CSV coordinates
- ❖ AI fuses evidence per location — it does not assume events

## HARDWARE REQUIREMENT

- ❖ No mandatory hardware — software-first prototype
- ❖ Future: IoT sensors, drones, satellite feeds, govt. systems

## CHALLENGES & MITIGATION

- ❖ Conflicting / duplicate reports → contradiction & embedding checks
- ❖ Outdated info → recency weighting | Scale → batch ingestion

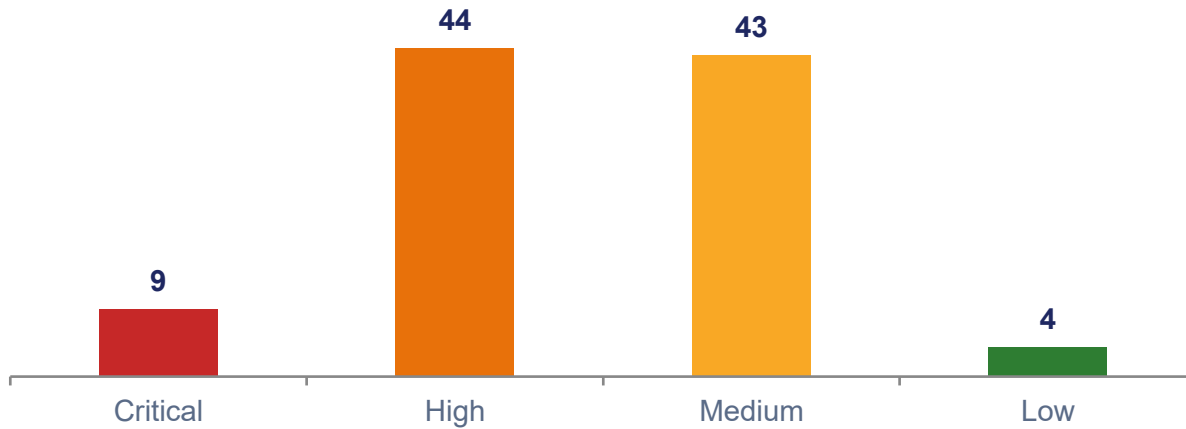


# PROTOTYPE DATASET & SCHEMA

## DATASET SCHEMA (SAMPLE FIELDS)

| Field                    | Example Value          |
|--------------------------|------------------------|
| report_id                | R001                   |
| location_zone            | Zone-E2 (Village A)    |
| disaster_type            | Flood                  |
| report_text              | “Flood reached school” |
| reported_need            | Rescue / Medical       |
| num_people_affected      | 350                    |
| source_type              | Field Worker Report    |
| source_reliability_score | 0.90                   |
| duplicate_report_count   | 11                     |
| verification_status      | Verified               |
| urgency_level (target)   | Critical               |

## SAMPLE DISTRIBUTION (100 SYNTHETIC RECORDS)



**Dataset type:** Synthetic + scenario-based (no real PII)

**Formats covered:** CSV/Excel, text, images (planned), location + time metadata

**Includes:** duplicate & contradictory reports, severity/priority labels

# IMPACT, INNOVATION & DIFFERENTIATION

## 1. Dynamic Evidence Fusion

Combines multiple independent sources instead of trusting one report

## 2. Explainable Confidence

Shows WHY a score is high: recency, confirmations, images, reliability

## 3. Temporal Intelligence

Detects Stable → Deteriorating → Critical trends over time

## 4. Uncertainty-Aware Priority

High risk + low confidence triggers  
“Verify this location first”

## 5. Dynamic Priority Ranking

Ranking updates automatically as new reports arrive

## 6. Resource-Aware Support

Recommends how limited resources should be distributed



# Thank You